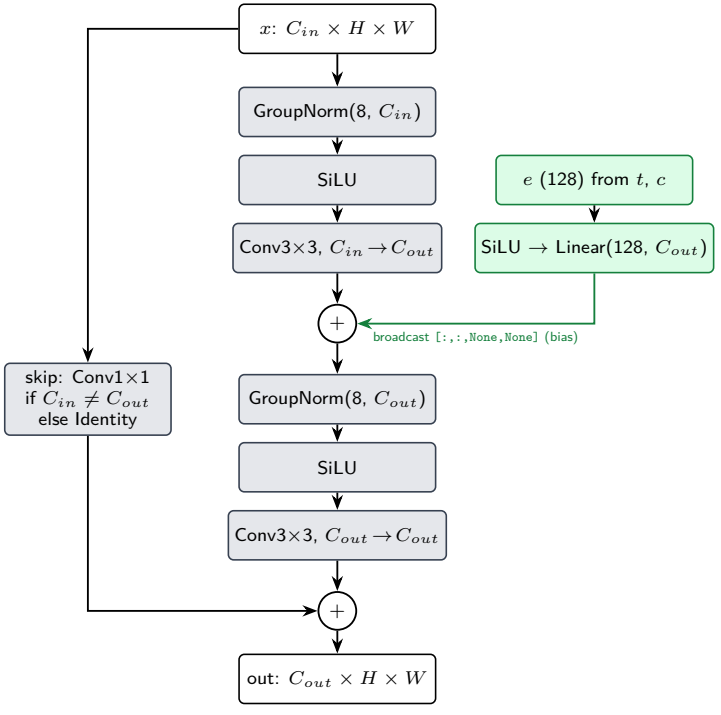
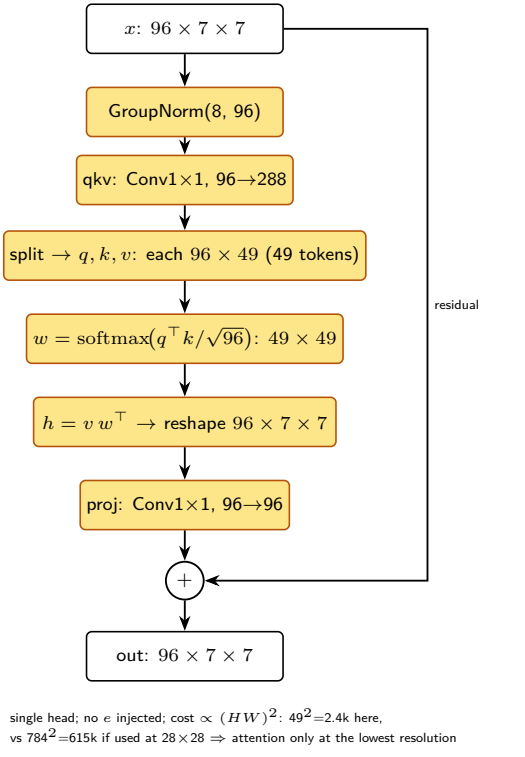

ResBlock($C_{in} \to C_{out}$) (code: ResBlock.forward)
x: $C_{in} \times H \times W$
GroupNorm(8, C_{in})
SiLU
Conv3×3, $C_{in} \to C_{out}$
e (128) from t, c
SiLU → Linear(128, C_{out})
broadcast [:,:,None,None] (bias)
skip: Conv1×1 if $C_{in} \neq C_{out}$ else Identity
GroupNorm(8, C_{out})
SiLU
Conv3×3, $C_{out} \to C_{out}$
out: $C_{out} \times H \times W$


Self-Attention at 7×7, $C = 96$ (code: Attn.forward)
x: $96 \times 7 \times 7$
GroupNorm(8, 96)
qkv: Conv1×1, 96→288
split → q, k, v: each 96×49 (49 tokens)
$w = \mathrm{softmax}(q^\top k/\sqrt{96})$: 49×49
$h = v\, w^\top$ → reshape $96 \times 7 \times 7$
proj: Conv1×1, 96→96
residual
out: $96 \times 7 \times 7$
single head; no e injected; cost $\propto (HW)^2$: 49^2=2.4k here,
vs 784^2=615k if used at 28×28 ⇒ attention only at the lowest resolution